

EDUCATION

- **The University of Michigan**
Computer Science B.S.E.

Ann Arbor, MI
09/2014 – 04/2018

EXPERIENCE

- **Michigan Medicine**
Software Engineer

Ann Arbor, MI
09/2023 – Present

- **Digital Pathology Tools:** Design, develop, and maintain the frontend, backend, and devops of a variety of tools, applications, and dashboards to improve digital pathology workflow using PHP, React, SQL, and Jenkins, increasing efficiency of labs, aiding in the digitization of pathology operations, and giving insights into process improvement opportunities. Currently leading the migration to a modern OpenShift stack using Laravel, TypeScript, and Postgres, improving scalability and separation of concerns by decoupling from our suite of lab applications
- **Virtual Slide Box Rewrite:** Modernized a legacy slide viewer by analyzing the existing PHP and JavaScript codebase, then rebuilt it from the ground up in Go, React, and Postgres using iisrv to serve high-magnification whole slide images. Deployed in OpenShift, improving scalability and maintainability
- **Whole Slide Image Tiledsource:** Contributed a custom tiledsource to OpenSeadragon, an open-source web-based viewer for high-resolution zoomable images, specifically geared toward improving image viewing performance in digital pathology
- **Sectra Integration & Support:** Partnered with key stakeholders and the Sectra team to deliver solutions including real-time image transfer and audit and an integrated information window in the desktop application, while providing ongoing support and troubleshooting to minimize downtime and ensure reliability
- **Slide Deidentification:** Overhauled a decades-old slide deidentification script using modern technologies and languages including Python, Docker, Jenkins, and Git to improve speed, reliability, and maintainability
- **Scanner Dashboard:** Developed a web application using PHP, React, and SQL to monitor and manage a fleet of high throughput slide scanners, including real-time status updates and error notifications, allowing for increased efficiency of lab operations and reduced scanner downtime
- **Internal Efficiency:** Create, maintain, and improve internal tools to address a variety of team and department needs, including data analysis, process automation, and workflow optimization. Improve manual and inefficient processes with robust, reliable, and maintainable scripts and applications using PHP, React, Python, Bash, SQL, and Jenkins

- **Coupa Software, Inc.**
Software Engineer

Ann Arbor, MI
01/2023 – 06/2023

- **Service Maintenance:** Analyzed and updated the Kafka custom connector, a critical platform component for loading data to and from JDBC-compatible databases. Refactored code using design patterns and recommendations from Sonar, increased unit test coverage from 20% to 80%, and improved documentation all while ensuring functionality was maintained

- **Nutshell, Inc.**
Software Engineer

Ann Arbor, MI
05/2022 – 11/2022

- **Feature Development:** Developed new features for the CRM, such as multifactor authentication, in PHP, GraphQL, and SQL using Docker, Git, Jenkins, and Visual Studio. Worked closely with the acquiring digital marketing firm to integrate our customer data and APIs with theirs, allowing for a more seamless transition for customers

- **Domino's Pizza, LLC**
Site Reliability/DevOps Engineer

Ann Arbor, MI
01/2020 – 05/2022

- **Kubernetes:** Implemented Pivotal Kubernetes Service (PKS) within the environment, which involved cluster buildout, environment configuration, monitoring, Helm chart development, and infrastructure-as-code
- **JenkinsX:** Built out JenkinsX within the PKS environment using Helm charts and running end-to-end validation scripts for each update to ensure the environment was stable and contained the latest changes, allowing for more rapid development
- **Java Installation Automation:** Developed a method for seamless and automated Java installation for runtime environments, cutting time spent by SRE resources installing Java on VMs. The change empowered development teams to more easily upgrade the JDK version for their own services, increasing the security and reliability of their applications

Technology Rotation Program

06/2018 – 12/2019

- **Digital Shoulder Surfing:** Working on a lean team as the sole backend developer, used Java, Spring framework, Maven, ActiveMQ and other tools to develop the backend service for a technology which enables in-store employees to see order information ahead of time, allowing them to prepare orders more quickly
- **Service Health Tool:** Developed a tool to check the health of all services, and to gather relevant data to ensure applications are running with the proper versions, saving time and finding issues by running during releases and datacenter changes. The tool automatically attempted to restart down services, improving the resiliency and maintainability of the environment

SKILLS

- **Programming:** Python, PHP, Go, Java, C++, C#, Bash, SQL, Postgres, R, Ansible, Puppet
- **Technologies:** Kubernetes, OpenShift, Docker, Helm, Jenkins, Nginx, .NET, Linux, Windows, Visual Studio, Git, GraphQL
- **Web Development:** React, HTML, CSS, JavaScript, TypeScript, Laravel, Axios, jQuery